Beyond Screening Off: Transition Closure as the Dynamical Criterion for Rainforest Admission

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Abstract

Scale-relative structural realism allows that real patterns may occupy more than one level of ontology, but the admittance criteria for those levels remain underdetermined. Franklin and Robertson (2024) offer the most developed recent proposal: emergence, understood as screening off plus novelty, earns an entity its place in the rainforest. This paper argues that screening off is necessary but not sufficient. A macro-variable can enter into robust and novel macrodependencies that render micro-details conditionally irrelevant, yet still fail to carry autonomous macro-dynamics of its own. What is missing is a transition criterion. A candidate macro-object qualifies when, for a fixed regime, horizon, and admissible intervention class, macrostate information is sufficient for macro-transitions, so within-class micro-differences do not change macro-level what-follows. In exact Markov settings, strong lumpability provides a benchmark realization. In non-ideal settings, closure is graded through leakiness and convergent diagnostics under fixed constraints. The interventionist machinery is compatible with structural realism because interventions are used as diagnostic probes for relational stability, not as metaphysical primitives. The result completes rather than replaces existing admittance proposals: Dennett's compression identifies candidate patterns, screening off and novelty provide a serious filter, and closure determines which surviving candidates carry genuinely autonomous transition structure.

Keywords: real patterns, rainforest realism, closure, strong lumpability, coarse-graining, structural realism, interventionism

1 Introduction: Closure and the Admittance Problem

Scale-relative structural realism says that the world is not exhausted by a single privileged descriptive level. The special sciences succeed by tracking higher-level patterns stable enough to support prediction, explanation, and intervention, and Ladyman and Ross's rainforest realism gave this thought its most influential metaphysical form: reality is populated by real patterns at many scales, not only by microphysical furniture (Ladyman and Ross 2007). But one question remains insufficiently settled. Which patterns earn admission into that rainforest, and which should be treated more cautiously as useful summaries, temporary proxies, or dynamically idle aggregates?

Franklin and Robertson offer the most developed recent answer (Franklin and Robertson 2024). Their proposal is that emergence earns rainforest admission when two conditions are met: the macro-variable screens off micro-details relative to the explanandum, and the macro-variable contributes something genuinely novel at that level. That is a substantial advance. It captures why obvious gerrymanders and trout-turkey style composites should be excluded, while preserving the thought that higher-level ontology is earned by stable macro-level dependence rather than by intuitive naturalness alone. They explicitly cast the issue as one of the "admittance criteria for the rainforest" (Franklin and Robertson 2024, p. 3), and they do so in a way meant to remain compatible with theoretical reducibility rather than to revive a strong irreducibility test.

Screening off is necessary but not sufficient. A macro-variable can satisfy observational screening-off conditions and enter into robust, law-like macrodependencies while still borrowing its apparent autonomy from upstream structure. What is missing is a transition criterion. A candidate macro-object qualifies when, for a fixed regime, horizon, and admissible intervention class, macrostate information is sufficient for macro-transitions, so within-class micro-differences do not change macro-level what-follows. Screening off shows that micro-details are conditionally irrelevant for a target dependence. Closure asks the further question whether the candidate carries its own transition structure under admissible perturbation. That further test also sharpens a pressure already implicit in rainforest realism itself: if a macro-candidate is to be more than a physically allowed summary, it must be non-redundant in a dynamical sense, not merely useful or compressive.

The dialectical structure is cumulative. Dennett's real patterns identify candidate structures through compression and projectibility (Dennett 1991); Franklin and Robertson add a serious admittance filter by requiring screening off plus novelty (Franklin and Robertson 2024); and closure adds one further step, distinguishing the surviving candidates that are genuinely autonomous from those that remain parasitic on hidden micro-identity or a shared upstream cause. The aim is to complete rainforest realism by supplying the missing dynamical downgrade condition.

Lewis observed that we have no principled reason to deny existence to the mereological sum of "the right half of my left shoe plus the Moon plus the sum of all Her Majesty's ear-rings," however sensible it is to ignore such things in ordinary thought (Lewis 1986, p. 213). Fair enough. But consider the difference between that composite and a hurricane. Knowing a hurricane's pressure organization and rotational structure tells you what it will do next without tracking every molecule. The shoe-moon composite carries no comparable transition structure; predicting its future requires independently tracking each component. Franklin and Robertson's criterion helps explain why the hurricane belongs in the rainforest while the shoe-moon composite does not. The argument here is that one can still construct macro-variables that survive screening-off tests while failing transition autonomy. The common-cause case in §3.3 is designed to show exactly that residual gap.

Throughout this paper, *gerrymandered* means dynamically incoherent: a partition that fails transition autonomy by requiring persistent within-class micro-bookkeeping. It does not mean merely intuitively disjunctive or aesthetically odd. By the same token, *parasitic* means that a macro-variable's apparent autonomy is borrowed from upstream structure rather than realized in the candidate partition's own transition organization.

The formal benchmark for this condition is strong lumpability in exact Markov settings. The non-ideal extension is graded closure via leakiness and convergent diagnostics under fixed constraints. The larger metaphysical payoff is that rainforest realism can then be read as a structured space of computationally closed coarse-grainings, not as a binary inventory of what is, or is not, real.

The argument is philosophical, not a methods paper in disguise. It provides a criterion and a disciplined protocol for applying it, while estimator selection, finite-sample behavior, and domain-specific implementation remain downstream methodological tasks.

1.1 Novelty and Positioning

The novelty claim has three parts.

1. Beyond screening off alone: Franklin and Robertson's criterion is retained as a necessary filter, but closure adds an intervention- indexed transition test that distinguishes autonomous macro-variables from parasitic ones. Their own account is valuable precisely because it frames rainforest admission without treating reducibility by itself as disqualifying (Franklin and Robertson 2024).
2. Beyond formal closure results alone: recent work distinguishes informational, causal, and computational closure as formal diagnostics (Rosas et al. 2024), but does not say when those diagnostics warrant ontological commitment or when commitment should be withdrawn. This paper supplies that missing philosophical layer: an objecthood criterion with explicit intervention indexing, graded verdicts, and downgrade conditions.
3. Beyond effective realism without downgrade discipline: recent work in ontic structural realism allows regime-dependent, scale-relative ontology (Ladyman and Lorenzetti 2024), but leaves open what operationally distinguishes robust admission from cases that should be downgraded or withheld. Closure supplies that discipline.

This is a conditional metaphysical proposal. Under structural realist and interventionist commitments, closure under admissible conditions is sufficient for robust macro-objecthood in regime. Readers who withhold those commitments can still accept a narrower conclusion: closure is at least a necessary anti-gerrymandering constraint on serious macro-ontology claims.

The positioning is deliberately dialogical. With Dennett, the paper keeps compression realism while refusing to let compression settle admittance by itself (Dennett 1991). With Franklin and Robertson, it agrees that screening off and novelty are indispensable but argues that they do not yet separate autonomous from parasitic macro-variables (Franklin and Robertson 2024). With Ladyman and Ross, it keeps the rainforest ambition while insisting on an explicit transition criterion for objecthood verdicts (Ladyman and Ross 2007). And with Ladyman and Lorenzetti, it treats interventions as diagnostics of structural autonomy rather than as metaphysical primitives (Ladyman and Lorenzetti 2024); their language of the scale relativity of ontology helps make clear why this does not collapse realism into convenience, since ontology can be regime-dependent while remaining mind-independent and answerable to objective modal structure (Ladyman and Lorenzetti 2024). One consequence of intervention indexing is that the closure criterion can diverge from purely observational diagnostics: a macro-variable can achieve predictive sufficiency by tracking a common cause rather than by instantiating autonomous transition structure (§3.3).

Several neighboring traditions are engaged where they become dialectically active: Simon and Wimsatt on near-decomposability (Simon 1962; Ando and Simon 1961; Wimsatt 2000), mechanistic accounts on organized explanation (Machamer et al. 2000; Craver 2007), Rosas et al. on formal closure diagnostics (Rosas et al. 2024), causal-emergence work (Hoel et al. 2013; Dewhurst 2021), and recent dynamic formulations of pattern realism (Millhouse 2022; Wallace 2024; Jiang 2024, 2025; Millhouse et al. 2026; Ladyman 2026). Each is addressed at the point where its specific contribution bears on the argument. The narrower claim is that even the strongest of these views leaves open a final admittance question that closure is proposed to answer.

1.2 Intervention, Structure, and the Causation Worry

One immediate tension should be stated early. Ladyman and Ross are skeptical that causation belongs in fundamental metaphysics, and recent effective ontic structural realism continues that broadly structuralist posture (Ladyman and Ross 2007; Ladyman and Lorenzetti 2024). By contrast, the criterion defended here is explicitly intervention-indexed. That can look like a retreat from structural realism to a Woodwardian metaphysics of causation. It is not.

Intervention does not function here as an ontological primitive. It functions as a diagnostic probe for structural autonomy. The claim is not that the world is fundamentally made of interventions, but that admissible intervention is our most disciplined test of whether a candidate macro-partition preserves stable relational organization under perturbation. What the criterion tracks is still structure. The intervention test asks whether that structure is autonomous at the candidate grain or whether the apparent autonomy was borrowed from elsewhere. Woodwardian intervention tests typically operate within an accepted variable scheme; the criterion here asks the prior question of when a candidate coarse-graining has earned the right to figure as a macro-variable at all.

The proposal is compatible with effective ontic structural realism because Ladyman and Lorenzetti explicitly allow ontology to be regime-dependent and scale-relative while remaining realist about the structures discovered there (Ladyman and Lorenzetti 2024). This paper agrees, but adds a sharper admittance condition: a scale-relative structure earns robust ontological standing when its transition profile is stable under the admissible perturbations that define the regime. In that sense, intervention is used here to supply operational discipline for effective realism rather than to compete with it. Their claim is not that ontology collapses into convenience, but that ontology is always effective ontology at some scale and under some modal profile (Ladyman and Lorenzetti 2024); this paper adds a stricter account of when such scale-relative commitment should be robust, qualified, or withheld.

To be clear: the paper requires intervention-invariance as evidence that a candidate partition tracks stable organization across physically real, regime-preserving perturbations to boundary conditions, couplings, or control channels, not interventions as metaphysical glue. That is a methodological route to structural stability, not a retreat to fundamental causal ontology.

1.3 Scope and Non-Claims

A preliminary distinction is needed. Closure can be stipulated or induced. In stipulated closure, autonomy is fixed by constitutive rules, as in chess or formal arithmetic: once the rules are set, transitions and consequences follow by construction. In induced closure, autonomy is discovered through dynamical screening-off in the physical world. This paper is concerned exclusively with induced closure in spatiotemporal systems. The paper offers no complete metaphysics of levels and does not settle questions about abstract objects or full mereology. Where those debates arise, they are handled only to protect the central claim from predictable misreadings. Still, the criterion is stated over state-transition structure: spatiotemporal dynamics are the paradigm case, not the only intelligible format in which closure questions can be posed.

Methodologically, the paper does not depend on the strongest anti-analytic rhetoric sometimes associated with naturalized metaphysics. The claim is narrower: closure is a usable philosophical constraint that can be motivated by dynamical and coarse-graining practice, whether or not one accepts every meta-metaphysical commitment in the broader *Every Thing Must Go* (ETMG) program (Ladyman and Ross 2007).

The contrast class is correspondingly focused. The paper rejects the view that compression and prediction are sufficient for realist commitment, but it also rejects a fallback to utility alone. The claim defended here is closure realism: compression and projectibility identify serious candidates, while robust objecthood requires transition autonomy under fixed regime, horizon, and admissible intervention class. Section 2 states the criterion and admissibility discipline. Section 3 gives the exact benchmark. Section 4 extends the same criterion to non-ideal settings and explains the graded verdict structure. Section 5 addresses the strongest objections. Section 6 summarizes the gains and limits.

2 Closure and Admissibility

2.1 Criterion in Plain Language

Closure can be stated without heavy formalism. A candidate macro-object passes when knowing its current macrostate is enough to determine its macro-future over a specified horizon and intervention class. If two microstates inside one macrostate produce different macro-transitions, closure fails at that grain.

Static patterns are not exceptions; they represent dynamical regimes where micro-fluctuations are screened off rather than absent, so persistence is itself a type of macro-transition that falls squarely within the criterion. What matters is transition sufficiency, not descriptive fit. A representation can summarize trajectories elegantly and still fail objecthood if it requires persistent within-class micro-bookkeeping to preserve forecast quality.

Since the formal machinery evaluates partitions of state spaces rather than physical things directly, an explicit bridge is needed. On the structural realist commitments defended below (§2.11), a macro-object just is stable relational and causal organization at the relevant grain; the partition identifies the object by identifying that organization.

A four-level distinction helps prevent recurring misunderstandings.

1. World dynamics: the implemented process itself.
2. Pattern type: stable transition structure supported by that process.
3. Pattern token: a concrete instance under specific boundary conditions.
4. Representation: a model, equation, classifier, or narrative that tracks the pattern.

A representation can fail while the pattern type remains real. A token can fail while the type remains robust. A token can also succeed while the type is weak, for example in narrow calibration windows. Closure claims in this paper are primarily type-level claims under specified regimes, then secondarily claims about token reliability under perturbation.

2.2 Formal Statement

Let micro-process be X_t and candidate macro-process be $Z_t = g(X_t)$. For horizon L, closure asks whether Z_t is sufficient for forecasting Z_{t+1}^{L} under a fixed regime and admissible intervention class.

The horizon object here is the L-step path distribution, not only one-step prediction. One-step tests can be used as diagnostics, but objecthood claims are indexed to the declared horizon target.

Informationally, the predictive side asks whether adding X_t beyond Z_t yields material gain for macro-future prediction. Interventionally, the causal side asks whether macro-transition laws remain stable across admissible perturbations without needing within-class micro-identification.

These are not competing tests. They are two readings of the same target: transition autonomy at the macro level.

2.3 Why This is More Than Compression

Compression is necessary but not sufficient for objecthood, because a compressed code can hide transition-relevant heterogeneity and still score well on narrow tasks. Closure blocks that loophole: does hidden heterogeneity matter for onward macro-behavior?

Closure therefore functions as an anti-gerrymandering condition, rewarding coarse-grainings that carry stable transition structure under declared constraints rather than convenient coding alone.

Put differently, this paper accepts Dennett's claim that compression success is evidence of objective patterning while denying that compression success is the full ontological test (Dennett 1991), and it accepts Ladyman and Ross's concern that real patterns should be projectible and structurally disciplined while adding an explicit transition criterion for adjudicating borderline cases (Ladyman and Ross 2007). The aim is not to replace those insights but to make their realist force less permissive.

This also answers the Haugeland-style pressure point: the transition-autonomy condition determines when pattern talk has ontological force rather than merely descriptive convenience (Haugeland 1998).

2.4 Closure and the Primacy of Physics Constraint

Ladyman and Ross's Primacy of Physics Constraint is often read negatively, as a warning against ontologies that float free of physics. But it contains a second, equally important pressure: macro-candidates must not only be physically compatible but must also avoid redundancy with lower-level description in the realism-relevant sense (Ladyman and Ross 2007). The missing question is what that non-redundancy amounts to once one gives up the idea that only the micro-level is fully real.

Closure gives that vague requirement operational content. A macro-candidate is non-redundant when micro-differences within one macroclass do not alter macro-transitions under the declared regime, horizon, and admissible intervention class. That is a stronger claim than descriptive usefulness, and a different claim from mere compression. A macro-description can be shorter, more tractable, and even explanatorily attractive while still remaining redundant in the sense that its apparent autonomy depends on continued micro-level repair.

The common-cause pressure case brings the point home. A proxy can look perfectly projectible under passive observation because upstream structure keeps its trajectories aligned, yet turn out to be transition-parasitic: perturb the candidate directly with an admissible intervention, and within-class micro-differences matter again. The apparent macro-autonomy disappears. Closure therefore sharpens the Primacy of Physics Constraint's non-redundancy demand by making it a question about transition structure rather than about descriptive convenience.

Read this way, closure is not an external supplement imposed on rainforest realism but a way of saying more exactly what it takes for a higher-level pattern to be physically respectable without collapsing back into lower-level bookkeeping. The realism-relevant discriminator is causal autonomy under admissible perturbation, not microphysical smallness by itself.

2.5 Admissibility and Hierarchical Evaluation

Closure is always relative to an intervention class, and the standard objection is that this invites circularity: who picks the class? The foundational answer is that the system's own transition dynamics do. A regime is a region of dynamical possibility carved out by the transitions themselves, and the admissible intervention class is the set of perturbations that regime tolerates without dissolving. The analyst's role is to read off that class honestly, not to invent it. Procedural discipline (below) ensures the reading is auditable, but the source of the constraint is the dynamics, not the analyst. Objecthood, on this view, is stability of macro-transition structure across the space of perturbations the regime actually sustains, not merely fit to observed trajectories.

Admissibility is fixed upstream of objecthood verdicts by three constraints:

1. Epistemic admissibility: interventions are measurable and implementable by bounded agents.
2. Dynamical admissibility: interventions preserve the target regime class.
3. Explanatory admissibility: interventions target variables with cross-context generalizability and non-trivial counterfactual reach, rather than one-off manipulations tuned to a single episode.

These constraints are physically anchored in available control channels and implementation structure, not in analyst preferences about favored ontologies. The explanatory constraint is set before closure verdicts and does not assume in advance which candidate partition will prove stable. This makes the interventionist notion of a "possible experiment" concrete and bounded, rather than leaving it at conceptual possibility where nearly any hypothetical manipulation might count (Woodward 2015). It also aligns with Woodward (2021)'s argument that correct variable choice requires satisfying "proportionality": minimizing the omission of dependency relations. Domain expertise still shapes regime identification and control-channel selection, but that shaping is declared, auditable, and separated from partition scoring.

Operationally, explanatory admissibility can be audited without looking at which partition wins. A candidate intervention class is stronger when it is implemented through control channels that already exist in the domain and are replicable across sites, operators, or runs. It is weaker when success depends on narrow, one-off tuning that fails under small, feasible perturbations of the same intervention family.

Pearl defends the $do(x)$ operator as an ideal "surgery" that can be defined mathematically regardless of physical feasibility (Pearl 2009). For causal inference, that idealization is productive: it lets one reason about effects even when direct manipulation is unavailable. But macro-objecthood, on the view defended here, requires tighter constraints than logical derivation. A partition that achieves closure only under physically unrealizable interventions, ones that no bounded agent could implement without destroying the target regime, does not support the kind of stable, intervention-guiding structure that warrants realist commitment. The admissibility constraints above ensure that closure is assessed relative to control channels that are actually available in the regime, not relative to arbitrary mathematical surgeries. The restriction concerns which of Pearl's outputs carry ontological weight, not the framework itself.

This point connects back to the foundational claim above. While we often speak of "interventions" as laboratory actions performed by human scientists, the admissible class is ultimately defined by the objective physical ecology of the system. A system's boundaries are constantly probed by naturally occurring environmental perturbations: thermal noise, chemical diffusion gradients, and mechanical impacts. A cell membrane objectively screens out certain molecular fluctuations whether a human experimenter is observing it or not. The transition dynamics of the cell carve out a regime, and that regime determines which perturbations are admissible: those the system processes or resists while maintaining its macro-transition structure. The intervention class is therefore not a human invention; it is a formalization of the space of possibility the dynamics already define.

Evaluation order matters.

1. Fix regime, horizon, admissible intervention class, diagnostics, and model class.
2. Compare candidate partitions under those fixed constraints.
3. If constraints are revised after inspecting results, restart and disclose the revision.

This order blocks back-fitting and makes disagreement tractable. If verdicts remain highly sensitive across nearby defensible admissibility specifications, commitment should be downgraded. Stated as a discipline, admissibility has two stages: first, intervention classes are fixed from independent physical constraints and established control practice in the target domain; second, closure is evaluated only relative to that fixed class.

2.6 Regime Individuation and Circularity

If regime individuation already commits to which macro-variables matter, the protocol risks circularity: closure testing presupposes regime identification, but regime identification might presuppose partition outcomes. The response is that defensible regime specifications draw on quantities characterizable without presupposing which candidate partition will succeed. Temporal parameters, energy scales, flow rates, and enforcement intensities are typically identifiable through physical indicators that competing candidate partitions can agree on before either is evaluated. In the traffic illustration, "weekday commuter traffic in one metropolitan corridor" is characterized by clock schedules, total vehicle counts, and road network topology, all quantities that investigators favoring either the lane-flow or vehicle-ID-cluster partition can agree on before either closure test is run. Regime specifications of this kind are anchored in lower-level or cross-cutting physical indicators rather than in the outcome of the very test they enable. Regime specifications should be grain-neutral across competing candidate partitions: a regime can itself be a macro-level description while still not presupposing which finer-grained partition will close within it.

The independence is imperfect in some domains, particularly biology and social science, where macro-variable choice and regime characterization can be entangled. The account treats that entanglement as a source of evidential weakness rather than a concealed defect. When alternative, independently motivated regime specifications produce divergent closure verdicts, the verdict ceiling should be downgraded to at most qualified. More broadly, regime individuation and partition evaluation are iterative rather than viciously circular. Initial regime characterizations are provisional, and closure testing can refine them. The process stabilizes into reflective equilibrium in the sense Goodman articulated: rules of inference and particular inferential judgments are justified by mutual coherence rather than by either serving as the unilateral foundation for the other (Goodman 1983). If iteration does not stabilize across nearby defensible starting specifications, the result should be treated as indeterminate for that regime rather than forced into a robust verdict. Here stabilization means that comparative ordering across independently specified candidate partitions stops shifting under small, defensible protocol variations; it does not presuppose one privileged grain in advance. The discipline comes from requiring that all revisions be disclosed and that each revision restart the evaluation rather than silently inheriting prior results.

2.7 Pattern Reality Versus Macro-Objecthood

A pattern can be real in a weaker sense without satisfying this paper's objecthood criterion. A representation can capture regularities that are descriptively useful while still failing closure under admissible interventions.

The criterion in this paper is stricter. It targets macro-objecthood, not any and every projectible summary. The paper can therefore preserve a generous attitude toward pattern detection while denying ontological standing to high-leak candidates.

That distinction also clarifies the paper's relation to Franklin and Robertson. Their screening-off condition is well suited to identifying when a macro-variable has become explanatorily and nomically serious (Franklin and Robertson 2024). The claim here is narrower and stricter: some variables that are serious enough to survive that filter still fail transition autonomy and so do not yet qualify for robust macro-objecthood.

The distinction also clarifies what this view contributes to composition debates. In van Inwagen's terms, the standing question is when some things compose a further thing (van Inwagen 1990). The present proposal does not offer a full mereology or a general answer to the Special Composition Question. It offers a narrower consequence in the paper's own register: for the restricted question at issue here, robust transition autonomy under declared constraints is enough to warrant composition claims in regime. That is a limited consequence for macro-object discourse, not a standalone solution to general composition debates.

So the dialectical burden is limited: the paper need only show that non-closed representations do not meet the objecthood standard defended here, not that they are worthless.

2.8 Admissibility Disputes

Admissibility disputes are expected, especially across domains. The adjudication rule is to compare candidate intervention classes by what they physically permit and whether they preserve the same target regime. Boundary-pressure perturbations can be admissible for storm dynamics, while molecule-by-molecule remote rewriting is not. In institutional settings, changing enforcement intensity can be admissible, while instant arbitrary rewriting of all agent commitments is not. The ontological consequence is direct. A storm satisfies closure under boundary perturbations because its pressure organization and rotational structure determine macro-transitions without tracking individual molecules. Under molecule-by-molecule rewriting, the same candidate fails, because admissibility now permits interventions that reach inside the macroclass and exploit within-class differences. The candidate has not changed; the admissible control channel has, and the verdict changes with it. Admissibility must therefore be declared and justified before scoring, not adjusted afterward to protect a preferred outcome.

This is a distinct issue from regime individuation and should be kept separate from it. Regime individuation asks which background dynamics define the target setting. Admissibility individuation asks which intervention classes count as ontologically relevant probes within that setting. The circularity worry is therefore different as well. A regime can be specified grain-neutrally while parties still disagree about what counts as explanatory reach inside that regime. The criterion answers that worry by tying explanatory admissibility to available control channels and implementation structure rather than to whichever interventions happen to vindicate a favored partition.

These constraints (existing control channels, cross-site replicability, and the exclusion of post hoc partition-protective manipulations) explain why admissibility is not secretly fixed by whichever partition wins.

In the special sciences, the front-end specification of regime, target, and admissible control channels is often interest-guided in a perfectly ordinary way. That does not make closure verdicts interest-relative once those specifications are fixed and openly defended. Where such specifications fail to stabilize across defensible alternatives, the right result is downgrade or indeterminacy rather than forced realism.

For transparency, three template classes are often useful as a starting grid: boundary nudges, coarse actuator controls, and policy levers. These templates are not universal, but they make admissibility comparison public and repeatable.

One guardrail is non-negotiable. Interventions that directly target preservation of partition labels, rather than system variables, are inadmissible because they trivialize closure by design. The test concerns whether transition structure is autonomous under physically meaningful controls, not whether labels can be protected by stipulation.

When two intervention classes are both admissible and target the same regime, the criterion permits plural testing rather than forced monism. Verdicts should report which admissibility class was used and how sensitive results are across nearby defensible classes. A candidate partition that appears closed only under one narrowly tuned class should be downgraded when sensitivity appears across nearby defensible alternatives.

The traffic example makes the point concrete. In commuter-corridor traffic, ramp metering and speed-limit controls count as admissible because they are physically available control channels in the target regime. Vehicle-by-vehicle remote rewriting does not count, not because it would embarrass a favored lane-flow partition, but because it is not part of the regime's actual control ecology. That is an implementation-level distinction, not a partition-presupposing one.

2.9 Canonical Criterion Statement

For ease of reference, the core criterion can be stated compactly:

A candidate coarse-graining $Z = g(X)$ qualifies for macro-objecthood in regime when, under fixed horizon L and admissible intervention class $\mathcal{I}$, macro-transition structure is autonomous up to declared tolerance, so within-class micro-differences do not materially improve macro-future prediction or intervention-guided control.

Candidate partitions must be specified independently of the particular sample trajectory used to score closure, for instance by a pre-declared construction rule or learning objective fixed before evaluation. Otherwise closure collapses into bespoke encoding, where any dataset can be made to look autonomous by post hoc recoding.

In exact Markov settings, strong lumpability is a sufficient benchmark realization of this criterion. In non-ideal settings, leakiness-centered diagnostics estimate distance from that ideal under fixed constraints.

2.10 Predictive and Interventional Closure

Predictive and interventional closure should be distinguished but not separated. Predictive closure concerns whether macrostate information screens off transition-relevant micro-detail for macro-future forecasting; interventional closure concerns whether macro-transition structure remains stable when the system is probed.

The evidential relation between them is asymmetric. Strong interventional closure typically implies predictive adequacy for the same target and horizon, while predictive adequacy alone can persist in cases where interventional stability fails under distribution shift. For that reason, the criterion treats predictive evidence as important but not final.

This asymmetry also clarifies burden of proof. Claims to robust macro-objecthood need either direct interventional support or compelling indirect evidence that tracks intervention-relevant invariance. Otherwise, the responsible verdict is qualified or indeterminate.

Indirect evidence can still be substantial: natural experiments, intervention-like policy shifts, quasi-experimental contrasts, and systematic failure under exogenous disruption can all reveal whether a candidate partition preserves its own transition structure across admissible changes. These sources do not collapse the distinction between predictive and interventional closure. They are useful precisely because they bear on intervention-relevant invariance rather than on in-sample fit alone.

The point matters especially in biology, economics, and the social sciences. In those domains, direct manipulation is often sparse, ethically constrained, or expensive. The ontological target nevertheless remains interventional closure. Where the evidence for that target is thin, robust verdicts should be rare. That is a feature of the graded verdict structure rather than a retreat to observational permissiveness.

2.11 Why Closure Carries Ontological Weight

A reviewer can still ask why transition autonomy should count as objecthood rather than as a mere success condition for modeling. The answer depends on the paper's explicit commitments. Under structural realism, what ontology should track is stable relational and causal organization rather than intrinsic micro-identity as such. Under interventionism, what counts as causally relevant structure is structure that supports stable manipulation and control.

This fits the regime-dependent, scale-relative orientation defended by effective ontic structural realism. The proposal does not ask for a return to one privileged level or to thick intrinsic objects, but for a sharper criterion of when a scale-relative structure has earned commitment, and when it should be downgraded, within a broadly structural realist framework (Ladyman and Lorenzetti 2024).

The meta-philosophical stance here follows what Woodward calls the methodological path to ontology: causal-interventionist tools are used to determine what earns realist commitment, rather than seeking independent metaphysical "truth-makers" or "grounds" that stand apart from the practices of manipulation and prediction (Woodward 2015). Scientific ontology, disciplined by closure, is the only ontology required for macro-objecthood verdicts. The demand for a further metaphysical foundation beyond stable dynamical autonomy under admissible intervention mistakes a philosophical habit for a substantive requirement.

A persistent worry is that prediction and interventional control are merely epistemic lenses, leaving the underlying ontology untouched. But this confuses the test with what is being tested. Prediction and intervention are diagnostic tools that reveal pre-existing statistical structure. The conditional independencies that allow a macrostate to screen off micro-details exist whether or not anyone computes them; they are the terrain, not the lens. Locality, finite signal propagation, and thermodynamic limits create these independencies as physical facts. Our predictive models succeed when they align with this objective causal topology, and fail when they do not. Closure is therefore an ontological criterion because it identifies where the world's phase space objectively decouples, not just where our minds prefer to chunk it.

Given those commitments, closure does not function as a convenient heuristic layered on top of ontology. It is the criterion that identifies when a candidate macrodescription tracks stable structure at the level where explanation and intervention are being assessed. This is why the view can remain compatible with microphysical completeness while still defending non-redundant macro commitment.

If one asks what the rainforest amounts to once admission is disciplined this way, the best answer on this view is not a loose plurality of levels but the ordered space of computationally closed coarse-grainings. Rosas et al. provide the formal architecture for talking about such relations among partitions (Rosas et al. 2024). The philosophical claim is that this ordered space gives the best interpretation of disciplined rainforest realism under the paper's stated commitments. Admission, qualification, and downgrade are therefore not floating labels imposed from outside the formalism. They locate a candidate within a structured space of more and less stable closure relations.

2.12 Structural Contrasts

Pearl's notion of modularity is the nearest structural contrast. Modularity is a property of a causal model whose variables have already been accepted; closure asks the prior question of whether a candidate coarse-graining has earned the right to appear as such a variable at all (Pearl 2009). Mechanistic accounts make the same point in a different idiom: organized entities and activities can support stable explanation, but that by itself does not show that a proposed macro-variable has been carved at the right place (Machamer et al. 2000; Craver 2007). Closure supplies that prior sorting question.

The proposal is also not vulnerable to Newman-style triviality pressure in structural realism. Closure is not extensional fit under redescription. It is modal and dynamical: arbitrary recodings typically fail once transition and intervention stability under declared constraints are required (Ainsworth 2009, pp. 141–145). The point, then, is not that a candidate is a merely "good variable" in some deflationary sense. It is that a closure-supporting partition tracks objective modal structure under admissible perturbation, whereas a merely convenient representation need not.

A related body of work on asymptotic explanation and universality classes bears directly on closure. Batterman argues that certain explanatory structures emerge at the interface between scales and resist reduction to any single descriptive level (Batterman 2002). That observation is compatible with this account: universality classes are precisely cases where heterogeneous micro-configurations yield the same macro-behavior, which is closure at the relevant grain. Batterman's focus is explanatory structure; closure adds an objecthood test that determines when such inter-scale structure warrants realist commitment rather than merely explanatory reliance.

The criterion therefore does not equate model performance with ontology. It asks whether the represented transition structure is autonomous, not whether one representation currently performs well.

2.13 Closure, Underdetermination, and Objecthood Discipline

One remaining concern is underdetermination. Even after the criterion is stated, a reviewer may argue that many distinct coarse-grainings can be tuned to look acceptable on available data. If so, closure might seem to collapse back into model choice rather than objecthood discipline.

The answer is to separate three questions that are often conflated.

1. Which candidate partitions are descriptively serviceable in a dataset?
2. Which candidates remain transition-autonomous under admissible perturbation?
3. Which of those candidates remain stable under modest horizon and admissibility variation?

Underdetermination is strongest at the first question and weaker at the second. It is often weakest at the third. Many candidates can fit observed trajectories. Far fewer preserve counterfactual structure when the regime is probed. Fewer still remain stable across nearby defensible setups. This staged filtering is exactly where closure earns its metaphysical role.

The account does not claim that every domain yields one uniquely privileged partition. It claims that objecthood commitments should be constrained by transition autonomy and robustness, rather than by fit alone. When underdetermination persists after those filters, the responsible result is qualified or plural commitment under transparent constraints, not forced monism and not unconstrained relativism.

Once admissibility, horizon, and intervention class are fixed, the question is not simply whether one favored partition closes. The question is which coarse-grainings close, which close only conditionally, and which fail. That answer is what the rainforest becomes under a closure discipline: the structured space of computationally closed coarse-grainings, ordered by refinement and by comparative stability. (Rosas et al. describe the formal result as a "lattice of nested computational structures" (Rosas et al. 2024, p. 2); whether the space has full lattice properties in every domain is an open formal question, but the ordering and nesting relations are what matter philosophically.)

When exact closure is unavailable, leakiness-centered diagnostics estimate comparative distance from closure, and commitment strength tracks the stability of that evidence.

3 Exact Benchmark: Strong Lumpability

Strong lumpability is introduced as a benchmark, not as destiny: a clean exact case in Markov settings where closure can be stated and checked without ambiguity.

Let partition cells under g be macroclasses. Strong lumpability holds when microstates within the same macroclass induce identical transition probabilities to all macroclasses (Kemeny and Snell 1960). When this condition holds, macro-transitions are autonomous by construction.

More precisely: for any two microstates x, x' in macroclass C_i, and any macroclass C_j, strong lumpability requires $\sum_{y \in C_j} P(y \mid x) = \sum_{y \in C_j} P(y \mid x')$.

Lemma. If the micro-process is first-order Markov and the partition induced by g is strongly lumpable, then the induced macro-process is itself Markov and transition-autonomous at the macro level.

Proof sketch (cf. Kemeny and Snell 1960, Theorem 6.3.2). Define a macro-kernel by summing micro-transition probabilities from any representative $x \in C_i$ into each macroclass C_j. Strong lumpability guarantees this sum is representative-independent. The macro-kernel is therefore well-defined, and the induced process over macroclasses is Markov with transitions given by that kernel.

A partition that satisfies this condition preserves transition structure at the macro grain. A partition that fails it mixes transition-heterogeneous microstates and therefore lacks macro autonomy.

Importantly, the Markov condition does not assume that history is irrelevant; it assumes only that the relevant state description has been fixed. In memory-bearing systems, that description may need to be history-enriched before the closure question is asked at all. That does not make non-Markovian lumpability easy, and it does not trivialize the formal difficulty. It does state the mathematically standard point that the closure test should be applied to the least state description that properly bounds the relevant memory.

Computational mechanics puts the same point in sharper language. For any macro-process Z, two prediction machines can be defined. The ε-machine is the minimal model that predicts Z's future from Z's own past, using only macro-level information.

The υ-machine is the minimal model that predicts Z's future from the full micro-past X, using everything available (Shalizi and Crutchfield 2001; Rosas et al. 2024). Closure holds when these two machines are equivalent: the ε-machine already captures everything the υ-machine knows about macro-futures, and extra micro-information is redundant for the macro target. This gives an exact ideal where closure is not approximate.

The limitation is equally clear: exact strong lumpability is uncommon in open, noisy, and path-dependent systems, which motivates the graded extension below without undermining the criterion itself.

3.1 Minimal Formal Illustration

The anti-gerrymandering role can be shown with a minimal symbolic example. Let microstates be $\{x_1, x_2, x_3, x_4\}$ with transition rows:

$$\begin{aligned}
P(x_1, \cdot) &= (\alpha, \beta, 0, 0), \\
P(x_2, \cdot) &= (\alpha, \beta, 0, 0), \\
P(x_3, \cdot) &= (0, 0, \gamma, \delta), \\
P(x_4, \cdot) &= (0, 0, \gamma, \delta),
\end{aligned}$$

with $\alpha + \beta = 1$ and $\gamma + \delta = 1$.

Partition A groups $\{x_1, x_2\}$ and $\{x_3, x_4\}$. Partition B groups $\{x_1, x_3\}$ and $\{x_2, x_4\}$. In A, members of each macroclass have matching onward profiles to macroclasses, so macro-transitions are autonomous. In B, members of one macroclass differ in onward profiles whenever $\alpha \neq \gamma$, so the macro-label hides a transition-relevant difference.

Partition B can still be made predictive by adding repeated within-class bookkeeping. A partition is *high-maintenance* when reliable macro-prediction repeatedly forces refinement of within-class distinctions, so that yesterday's macrostate labels must be supplemented by new micro-bookkeeping to retain accuracy. That is exactly the case this paper excludes from robust macro-objecthood. On a fixed sample trajectory, this bookkeeping can make Partition B look nearly as predictive as Partition A. The difference appears under admissible perturbation: A preserves macro-transition structure without renewed microstate tracking, whereas B does not.

Both partitions are definable. Only one preserves transition autonomy. Closure therefore discriminates between legitimate macro-candidates and merely codable aggregates.

3.2 If an Unusual Partition Closes

A common reaction is that a strange disjunctive partition might satisfy the formal condition in a symmetric system. That is correct. On this account, if such a partition supports autonomous macro-transitions under fixed constraints, it counts as real at that grain.

That is not a concession to arbitrariness; the criterion tracks transition coherence, not intuitive naturalness. If one wants a separate naturalness filter, that is an additional criterion and should be declared as such.

That result should also be read alongside Andersen's argument that structure may be dense rather than sparse (Andersen 2025). If multiple partitions close at different grains, that is not by itself an embarrassment for the view. It is what one should expect if the world's structural organization is richer than any single privileged carving. What matters is whether the candidate partition earns admission by closure under stated constraints, not whether it is the only possible closed partition.

The exclusion claim should therefore be read carefully. The criterion excludes dynamically incoherent, high-leak aggregates. It does not exclude every unusual grouping.

3.3 What the Formal Machinery Alone Does Not Provide

The mathematical property and the philosophical criterion are different things. Four elements of this proposal have no counterpart in the formal literature alone.

First, intervention indexing. Causal-state constructions classify histories by equivalence of future distributions under the observed process, which is primarily an observational-predictive equivalence relation (Shalizi and Crutchfield 2001). The criterion here adds an admissible intervention class as a parameter of the closure test. Objecthood claims are indexed to what the system does under perturbation, not only under passive observation, which separates the criterion from a purely statistical diagnostic. Rosas et al.'s framework distinguishes informational, causal, and computational closure as complementary formal diagnostics (Rosas et al. 2024). This paper treats predictive sufficiency as evidential, but ties objecthood commitment to admissible-intervention stability and to the absence of hidden micro-identity bookkeeping.

Second, an explicit ontological interpretation. Strong lumpability tells you when a coarse-graining preserves transition structure. It does not tell you whether that preservation warrants realist commitment. The philosophical work is to argue that, under stated commitments, transition autonomy under admissible interventions is sufficient for macro-objecthood, and to give the conditions under which that claim should be downgraded or withdrawn.

Third, a graded commitment protocol. The formal literature offers exact conditions and, more recently, approximate-closure measures. It does not supply a protocol for translating those measures into warranted ontological verdicts with explicit robustness requirements and downgrade rules. The selective commitment structure (robust, qualified, indeterminate) is a philosophical addition. These are not merely epistemic confidence labels. They report where a candidate sits in the space of more and less stable closure relations: robust where closure survives modest admissibility and robustness checks, qualified where closure is real but regime-fragile or only partially secured, and indeterminate where the candidate's place in that structure is not yet stable enough to underwrite commitment.

Fourth, an anti-gerrymandering argument. The argument that closure is the right anti-gerrymandering condition, and that compression without closure is insufficient for objecthood, because compression can be won by codings that effectively preserve hidden micro-identities, is not a theorem. It is a philosophical claim defended by the structure of Sections 2 through 4. The formal illustrations do not independently prove that claim. They make the relevant failure mode transparent enough for the philosophical criterion to be assessed directly.

A concrete case shows the divergence. Suppose a composite variable aggregates two subsystems that are both driven by a shared upstream oscillator. Under passive observation, the composite's macro-state predicts its own future as well as the full micro-state does, because the shared driver synchronizes the subsystems and makes their joint trajectory compressible. An ε-machine built on the composite would find no residual micro-information worth adding. Yet if an admissible intervention targets the composite directly, say by perturbing one subsystem while leaving the driver intact, the previously synchronized trajectories decouple and the composite's macro-transition structure breaks down. The predictive sufficiency was parasitic on the common cause, not grounded in autonomous transition organization within the composite itself. Observational diagnostics alone cannot distinguish this case from genuine closure; the intervention-indexed criterion can.

This case creates a wedge against the screening-off-plus-novelty package. Franklin and Robertson's criterion asks whether the macro-variable enters into a genuine macrodependency that renders some lower-level detail conditionally irrelevant while contributing something novel at the macro level (Franklin and Robertson 2024, p. 1). The common-cause proxy satisfies each component of that package under passive observation. It is unconditionally relevant because its states track target-relevant differences. It is conditionally irrelevant relative to the target explanandum because, once the proxy state is fixed, further micro-differences in the driven subsystems add no further predictive leverage within the observed regime. And it has macro-level novelty because the dependence is captured at the proxy grain rather than by listing the underlying microconfiguration. Under passive observation, the shared driver can sustain this conditional independence dynamically across time. The case therefore deserves to be treated as a genuine pressure case for the screening-off-plus-novelty package rather than as a verbal trick or a simple failure of diagnosis.

Closure addresses that remaining question. If an admissible intervention perturbs the proxy while leaving the common cause intact, the apparent autonomy disappears. The macro-variable was not organizing its own transitions; it was borrowing its stability from upstream structure. Formally, that is what parasitism amounts to here: predictive or screening-off success depends on upstream organization in a way that the candidate partition does not preserve under direct admissible perturbation. What the case shows, then, is not that screening off is spurious or dispensable, but that screening off by itself does not yet settle the issue of autonomous macro-objecthood.

The intervention test here is accordingly not just standard Woodwardian causal testing in new language. Standard interventionist analysis asks whether one variable causes another within an already accepted partition. The question here is prior to that: it asks whether a candidate partition itself carries autonomous transition structure under admissible perturbation.

The structured-space payoff (§2.13) has three further consequences. First, levels on this view are discovered features of system dynamics rather than metaphors or analyst conveniences. Closure identifies which partitions track autonomous transition structure under declared constraints, so the level structure is read from the dynamics rather than stipulated in advance. Second, levels relate by nesting rather than by elimination. When closed grains stand in refinement relations, the relevant metaphysical relation is ordered inclusion within a structured space of coarse-grainings, not the disappearance of one level into another. Third, the micro-level is not privileged simply by being smallest. Microphysics remains the implementation layer, but what matters for rainforest admission is causal autonomy under admissible perturbation, not microphysical size by itself. Rosas et al. provide the formal architecture for this ordered space; the claim that membership tracks robust ontological seriousness is this paper's philosophical contribution.

A similar point applies to causal-emergence work. Effective-information analyses identify when macro-descriptions gain determinism relative to micro-descriptions (Hoel et al. 2013); Dewhurst (2021) argues that such gains support only epistemic emergence absent further constraints. That diagnosis is correct as far as it goes, but it identifies a gap rather than filling it. A partition can increase effective information while remaining fragile under intervention or horizon variation. The approach here treats such gains as corroborating evidence within a closure-governed criterion, not as a standalone ontology test.

Relatedly, Wallace's dynamic treatment of higher-level ontology is right to push the debate away from mere collections of parts and toward structure "in the distributions and dynamics" (Wallace 2024). But a commuting or phase-space friendly representation is still not yet a full admittance criterion. Closure asks when the higher-level structure holds up under admissible perturbation rather than merely being calibrated to observed trajectories. Ladyman's recent contribution to the real-patterns volume moves in the same dynamic direction. He characterizes real patterns in terms of "projectible generalisations that predict and explain" (Ladyman 2026, p. 1). The proposal here is that what remains to be added is explicit admission and downgrade discipline.

So the paper is not computational mechanics or causal-emergence analysis with new labels. It is a philosophical criterion that borrows formal tools while adding intervention-indexed objecthood conditions, an explicit commitment protocol, and a sustained argument about why closure is the right fix for pattern-realism's permissiveness problem.

4 Approximate Closure Without Instrumentalist Drift

4.1 Why Approximation is Expected

In complex systems, boundaries leak and couplings shift across regimes. Exact closure is therefore unusual. A credible realism criterion cannot require perfect closure everywhere. The study of approximate aggregation has a long history, especially Simon's account of hierarchy and near-decomposability, where strong within-subsystem couplings can coexist with weaker cross-subsystem couplings and thereby separate short-run local dynamics from longer-run aggregate behavior (Simon 1962; Ando and Simon 1961). The extension here is philosophical rather than technical: it uses graded closure to support graded ontological verdicts, not to improve estimation. Rosas et al.'s distinction between informational, causal, and computational closure is useful here because it shows why the formal story is already graded and plural at the diagnostic level even when this paper uses it to motivate one ontological criterion (Rosas et al. 2024).

Approximation is the expected non-ideal form of the same criterion, provided constraints are fixed and diagnostics are comparative.

Ontology tests should be type-level before token-level. A pattern type can be robustly closed for a regime even when a specific token fails because of boundary violation, atypical perturbation, or timescale mismatch. One anomalous token is therefore not decisive. The relevant question is whether failures are exceptional or systematic for the type under the stated constraints. Systematic failure, where within-class micro-differences repeatedly matter for macro-transitions across tokens under fixed constraints, is what triggers type-level downgrade rather than token-level exception.

4.2 Leakiness as Canonical Target

Leakiness measures how much within-class micro-detail still improves macro-future prediction once the current macrostate is fixed. A canonical quantity is the conditional mutual information $I(X_t; Z_{t+1} \mid Z_t)$ (Cover and Thomas 2006, eqs. (2.60)–(2.61)): low values support closure, high values indicate hidden transition-relevant heterogeneity. In plain terms, a non-zero value means that residual microdetail still helps predict the next macrostate even after the present one is fixed. That is the signature of a leaky partition.

Using information-theoretic measures to define ontology might appear to assume a computationalist metaphysics, but the justification is strictly physical. Information processing incurs thermodynamic costs (Landauer 1961). A system, whether a biological organism or an engineered control mechanism, that tracks full micro-details without gaining macro-predictive advantage incurs unsustainable energetic overhead. Evolution and physical self-organization therefore robustly select against leaky boundaries.

Measuring leakiness is not an imposition of computer-science metaphors onto nature; it is a formal accounting of the physical and thermodynamic constraints that force systems to dimensionally reduce their own state spaces. A low-leakage partition tracks the precise joints where the physical world successfully minimizes this metabolic and computational cost.

In this paper, leakiness is the default target quantity. Other diagnostics, such as within-class transition divergence and predictive gain from added micro-features, function as estimators or proxies when direct estimation is limited.

Leakiness should therefore be read comparatively and through robustness checks, not as a context-free absolute cutoff. Absolute tolerance is domain-relative, but high sensitivity of leakiness rankings to modest defensible modeling choices is itself a downgrade trigger.

Comparative ranking is necessary but not sufficient, since a candidate can beat nearby alternatives and still fail objecthood if absolute leakiness remains high and instability persists under modest robustness checks. Both relative superiority and minimum viability are required.

The viability floor can be stated without a numeric threshold. A partition fails minimum viability when, across the declared robustness checks, within-class micro-features yield consistent, non-negligible gains in macro-future prediction or intervention response relative to the best macro-only model. Persistence of that pattern across diagnostics and perturbation tests is decisive: the partition has not achieved the transition autonomy the criterion requires, regardless of how it ranks against competitors.

The term "non-negligible" can be given a comparative anchor without fixing a universal numeric threshold. Leakiness is most clearly non-negligible when adding within-class micro-features yields prediction gains comparable in magnitude to the gains the macro-partition itself provides over a naive baseline model. If the macro-partition delivers substantial predictive improvement over no partition at all, and within-class micro-features add only marginal further gain, that residual leakiness is negligible for most objecthood purposes. If micro-features instead add gains approaching or matching the macro-partition's own contribution, the partition is not doing most of the relevant work, and minimum viability is not met. This framing makes the judgment about a single tractable quantity, namely relative predictive contribution, rather than an abstract absolute cutoff, and it makes cross-domain comparisons more tractable. The two requirements work in sequence: the viability floor is applied first as a minimum gate, excluding candidates regardless of how they compare to rivals, while comparative ranking then assigns verdict categories among candidates that pass it. To keep this anchor non-discretionary, the baseline model class and gain metric must be fixed upstream with regime and admissibility, and post hoc revisions trigger restart and disclosure.

Approximate closure is therefore not a discretionary replacement for exact closure. It is the non-ideal evidential route to the same ontological target. Judgment is constrained by upstream fixing of regime, admissibility, baseline model class, and gain metric, by comparative stability across modest robustness checks, and by explicit downgrade or indeterminacy when those rankings do not stabilize.

Verdict-category boundaries carry genuine vagueness because the underlying property of transition autonomy is itself graded. This is a feature rather than a defect: biological fitness is real and explanatorily significant without admitting a sharp boundary between fit and unfit. What matters primarily is ordering stability. When partition A consistently shows lower leakiness and greater perturbation stability than partition B across diagnostics, that ordering should be robust even if the exact placement of the boundary between "qualified" and "robust" carries some domain-conventional element. A partition that dominates alternatives across diagnostics and remains stable under modest changes to horizon, intervention distribution, and robustness checks earns the stronger verdict regardless of where one draws the categorical line.

4.3 Procedural Safeguards in Non-Ideal Cases

Approximate closure claims require explicit safeguards.

1. Fix regime, horizon, admissibility class, and diagnostics before comparative scoring.
2. Compare candidate partitions under the same fixed setup.
3. Test robustness under modest changes in horizon and intervention distribution.
4. Apply a minimum-viability floor: if all candidates remain high-leak and fragile, return no robust objecthood verdict for that regime.
5. Report verdicts as robust, qualified, or indeterminate.

These safeguards also fit Wimsatt's robustness-analysis idea: confidence should rise when partially independent ways of detecting the same structure converge, and it should fall when apparent agreement is driven by shared assumptions or disappears under small perturbation (Wimsatt 1981). Multiple diagnostics do not replace criterion-level argument with a vote count. They reduce the risk that a favorable verdict is an artifact of one estimator, one horizon choice, or one admissibility specification.

This keeps threshold choice procedural rather than discretionary. It also addresses a predictable confusion. Estimation difficulty is an epistemic limit on access, not a defect in the metaphysical criterion itself. What is graded in non-ideal cases is not confidence alone, but the stability of closure itself across regime, horizon, admissibility, and robustness variation.

Selection discipline. One concern from both reviewers and readers is that flexibility can silently re-enter through upstream choices. The answer is a single discipline rule: admissibility, state construction, horizon, and diagnostics are fixed before scoring, and post hoc revisions trigger restart and disclosure. The point is to make judgment auditable, not to eliminate it, and to prevents favored partitions from being protected by moving criteria.

4.4 Regime Dependence and Projectibility

Regime dependence is a claim about actual transition structure under specified constraints, not about observer-relativity. A pattern can be closed in one regime and leaky in another because the structure has changed, or because the available control channels have shifted. Since the admissible intervention class is defined by the regime's transition structure (§2.5), a change in that structure can alter both the admissible class and the closure verdict even when the system's micro-dynamics remain the same.

The metaphysical stance is explicit: regime and horizon index the modal profile being claimed, not a concession that anything goes.

An ordinary physical analogy helps. Ice crystals are objective structures only within certain temperature and pressure regimes. That regime dependence shows that objective structure can be conditional on equally objective background constraints without becoming epistemically subjective.

Projectibility provides a further check. Defensible regime specifications should support stable induction under modest counterfactual variation. Narrowly engineered regimes that protect a favored partition usually fail under small context shifts. When that occurs, the candidate was never robustly closed in the relevant sense.

Here the paper is closest to Ladyman and Ross. Projectibility is not treated as an optional methodological virtue. It functions as a realism-relevant stress test on whether a closure claim survives beyond calibration conditions (Ladyman and Ross 2007). Closure without projectible robustness is too cheap to support robust objecthood claims.

But projectibility is still not enough by itself, because a pattern can remain projectible while borrowing its stability from upstream structure (see §3.3). Closure asks the further question of whether the candidate partition carries its own transition structure under admissible perturbation.

Ladyman and Lorenzetti's effective ontic structural realism sharpens the same point at the metaphysical level. If ontology is regime-dependent and scale-relative, then the question is not whether a candidate belongs to a single final inventory of being, but whether it is stable enough in the relevant regime to merit commitment there (Ladyman and Lorenzetti 2024). Closure is offered here as the operational discipline for making that judgment explicit.

Regime dependence itself should be split into two forms. Ontic regime dependence occurs when system structure changes, such as phase transitions or boundary-condition shifts. Epistemic regime dependence occurs when measurement or control access changes while underlying structure remains fixed. The first changes what is there to be tracked. The second changes what can be warranted from available evidence.

4.5 One Distributed Illustration

Macro does not mean large or spatially contiguous. Coarse-graining can be logical and distributed. Monetary systems illustrate this point without requiring new machinery. A clarification is needed, given the earlier restriction to induced closure (§1.3). The relevant closure here is not the stipulated rules of the financial system (those are constitutive), but whether the physical and institutional infrastructure that implements those rules sustains autonomous macro-transitions under admissible perturbation. The failed-implementation case below shows exactly where induced closure breaks down even when the stipulated rules persist.

The macro-transition structure of transactions can remain stable across heterogeneous micro-realizations, such as cash tokens, ledger records, and digital balances, under admissible legal and financial interventions. If that stability holds, macro-objecthood is warranted in regime. If implementation channels degrade, closure can fail even while symbolic representations persist.

A minimal failure case clarifies the point. A state can retain formal legal tender rules while losing reliable payment enforcement and settlement implementation. Closure fails not because the rules changed but because the physical infrastructure that enforced screening-off is gone. The representation persists but transition autonomy degrades, and closure-based commitment should be downgraded. The same structure applies more broadly: the criterion explains when and why a social entity stops being a real macro-object without requiring the claim that it was never real, or that it remains real simply because people believe in it. On the broader rainforest realist picture, the relevant issue is whether transactional organization still supports the economic structure in question, not whether the label survives institutionally (Ladyman and Ross 2007).

That is also the point of contact with the Ross-inflected side of rainforest realism about institutional kinds. Institutional ontology need not be arbitrary to be fragile. What closure adds is a way of explaining why some institutional kinds retain robust standing under heterogeneous realizations and admissible interventions, while others must be downgraded once enforcement, settlement, or trust structure no longer sustains the relevant macro-transitions (Ross 1995).

Organisms give a complementary illustration. In many physiological regimes, membrane and regulatory organization screen off large amounts of molecular variation for the target transitions, so intervention on organ-level variables remains predictively and manipulatively effective. When those screening structures fail, for instance under severe systemic breakdown, the same macro-description can become leaky and require finer-grained tracking. The point is not that organism talk is always closed. The point is that closure can be regime-robust without being regime-universal.

Across these illustrations, the verdict structure does real discriminative work. The monetary example in a well-functioning system approaches robust objecthood in the declared regime: macro-transition structure remains stable across heterogeneous realizations and admissible interventions. The organism example falls into the qualified range: within established physiological regimes, membrane organization and regulatory control screen off large amounts of molecular variation, making organ-level interventions reliably effective, but leakage appears at ecological or pathological boundaries where those screening structures degrade. The failed-implementation case represents a downgrade: formal representations persist while transition autonomy erodes, warranting withdrawal of robust commitment. The three cases illustrate the verdict categories in operation, not as abstract labels but as outcomes determined by the same diagnostic criteria applied to different evidential situations.

The illustration does one job only. It shows that closure tracks transition structure, not spatial shape. Nothing in the argument depends on taking a stance on broader social ontology.

Taken together, these cases also show what the criterion is expected to include, qualify, and exclude. It includes candidates such as stable monetary systems when heterogeneous realizers still preserve the same macro-transitions under admissible intervention. It supports qualified commitment in regime-fragile cases such as organisms, traffic partitions, or other wholes whose screening structures hold only across bounded horizons or regimes. And it excludes or downgrades cases whose apparent stability is merely borrowed, such as projectible proxies, gerrymandered composites, or failed posits whose transition autonomy collapses once the supporting structure is perturbed or removed.

4.6 Failure Conditions and Downgrade Rules

The account should also say clearly when commitment should be withdrawn or downgraded. Four failure patterns are especially relevant.

1. Persistent high leakiness across defensible diagnostics under fixed constraints.
2. Strong disagreement among diagnostics that does not resolve under modest robustness checks.

3. High sensitivity of verdicts to small, defensible changes in admissibility or horizon.
4. Candidate sets where no partition clears the minimum-viability floor.

When these patterns appear, the right response is not forced binary judgment. The right response is explicit downgrade to qualified or indeterminate status. This keeps the criterion resilient without pretending that every case must produce a sharp ontology verdict.

Conversely, when closure persists across distinct, independently motivated admissibility classes within the same regime, that cross-admissibility stability raises evidential confidence for robust commitment.

Type-token reminder: a downgraded token does not automatically refute type-level closure. The key question is whether instability is exceptional at token level or systematic for the type under the stated constraints.

4.7 Stable Versus Merely Entailed Patterns

The graded account also supports an important distinction. A pattern can be entailed by microhistory and laws without being stable as a macro-handle. Entailment alone is cheap; stability under admissible perturbation is demanding. The distinction matters for permissiveness debates, because a contrived disjunctive construction can be true of a realized trajectory while still failing closure: it can require continual within-class micro repair, fail under modest regime shifts, and lose interventional reliability.

A historical case makes the pattern concrete. Phlogiston theory compressed combustion phenomena under a single macro-variable, but the partition required persistent bookkeeping to survive: when metals gained weight upon burning, theorists introduced "negative phlogiston"; when different substances showed different weight changes, further ad hoc parameters appeared. Each patch was a new within-class distinction imported to preserve macro-prediction. By contrast, oxygen theory required substantially less such repair: the macro-variable (oxidation state) screened off the relevant chemistry without persistent bookkeeping. The critique here is structural, not retrospective ridicule. A candidate that continually imports corrections to preserve macro-prediction behaves like a non-autonomous partition, and the closure framework detects this failure directly.

So gerrymandered constructions are not necessarily false; they usually fail to qualify as macro-objects. They may remain descriptions, but not object-level descriptions in the relevant regime. The same applies to computationally adequate models more broadly. A higher-level model can be useful for prediction or control without meeting closure standards. Computational adequacy means the model serves some purpose; closure requires that transition-relevant micro-differences are screened off in the specified regime. A computationally adequate but high-leak representation remains a valuable tool, but it does not automatically earn macro-object status.

4.8 Practical Qualification Under Sparse Intervention Access

In many domains, direct interventions are sparse, ethically constrained, or expensive. Sparse intervention access changes evidential strategy, not the criterion itself.

The evidential order is asymmetric: predictive success can raise confidence, but only interventional stability can underwrite objecthood commitment.

Where intervention data are limited, predictive closure functions as an operational indicator while interventional closure remains the ontological target. A model can fit historical trajectories and still fail under novel perturbation. For that reason, claims should be qualified by available intervention access and downgraded when out-of-regime behavior is unstable.

In-sample prediction is therefore not treated as decisive. The standard remains robustness under admissible perturbation, even when that robustness must be assessed indirectly.

Relevant evidence need not come only from humanly executable interventions. In observational sciences it can also come from naturally occurring disturbances, comparative cases, exogenous shocks, policy changes, and other perturbations that bear on the same transition structure. Such indirect intervention evidence often reveals whether a macro-description remains stable when the system is pushed off its ordinary trajectory.

This point also explains why diagnostics should be triangulated. Predictive closure can look strong in-sample while interventional closure fails under admissible perturbation. Partial observability can hide within-class heterogeneity that later appears as instability. Nonstationarity can make a previously low-leak partition unstable outside its calibration window. Triangulation does not remove these risks, but it makes them visible.

The monetary illustration is a useful case. A well-functioning payment system can look observationally stable for long stretches. What matters for stronger commitment is how that stability behaves under settlement delays, enforcement shocks, trust disruptions, or policy changes that function as admissible perturbations. Where those perturbations leave transactional organization intact, robust verdicts become plausible. Where the same structure survives only observationally, or collapses once such pressures appear, the right verdict is qualified rather than robust.

4.9 Conceptual Contrast: Near-Tie Prediction, Different Ontology Verdicts

Two partitions can show similar short-horizon observational performance and still receive different closure verdicts, and this is where the criterion does its real philosophical work.

Suppose partition P_1 and partition P_2 produce comparable one-step observational fit in calibration data. Under fixed admissible interventions, P_1 preserves stable macro-transition structure while P_2 requires repeated within-class micro refinements to maintain performance. Observationally, they may look close. Structurally, they are not.

On this account, P_1 receives the stronger objecthood verdict because it remains transition-autonomous under the fixed constraints. P_2 can remain useful as a representation, but its dependence on recurring hidden repair counts against macro-object standing.

A concrete sketch helps. In traffic modeling, one partition may track density and flow by lane segment. A rival partition may track arbitrary vehicle-ID clusters that happen to fit one week of observations. In-sample one-step fit can be similar. Under admissible perturbations, such as ramp metering changes and speed-limit adjustments, the lane-flow partition remains stable while the ID-cluster partition becomes brittle. Under modest horizon extension, leakiness for the ID-cluster partition rises sharply. The verdict is then robust or qualified objecthood for the lane-flow partition and downgrade for the rival.

Under the same disclosure discipline, the contrast can be stated compactly:

1. Target partitions: P_{lane} (lane-segment density and flow) versus P_{id} (vehicle-ID clusters).
2. Regime and horizon: weekday commuter traffic in one metropolitan corridor; horizon set to short-run control windows and one modest extension beyond calibration.
3. Admissibility class: control channels that traffic operators actually use in that regime (ramp metering and speed-limit controls), excluding interventions that require implausible vehicle-level actuation.
4. Diagnostics: predictive gain from within-class microfeatures, within-class transition-profile divergence, and intervention response invariance under admissible perturbations.
5. Verdict: P_{lane} remains comparatively stable and earns robust or at least qualified objecthood in the declared regime; P_{id} remains instrumentally useful in calibration but is downgraded once perturbation and modest horizon extension are applied.

A second illustration shows the criterion sorting between two scientifically rigorous partitions. Consider two ways to divide an organism into macroscopic parts. One partition, P_{organ}, follows functional physiological boundaries: heart, lungs, liver. A rival partition, P_{slice}, divides the body into strict transverse geometric planes, as used in medical tomography. Under resting observation, both achieve comparable short-horizon compression of local mass, fluid density, and electrical activity. Under an admissible physical perturbation, such as cardiovascular drug administration, electrical pacing, or sudden exertion, their transition autonomy diverges. P_{organ} preserves its macro-transition structure: cardiac electrophysiology has well-characterized macro-transitions (sinus rhythm, tachycardia, fibrillation) that are predictable from organ-level variables without tracking individual ion channel states, and the admissible intervention class (pharmacological, electrical) operates at precisely that grain. P_{slice} fractures, because functional tissues span across geometric planes; predicting the macro-future of a given slice now requires tracking the specific intercellular signals and fluid volumes crossing its boundary. On this view, P_{organ} earns robust objecthood in the physiological regime because it isolates transition-autonomous parts; P_{slice} remains a valuable diagnostic tool but its recurring dependence on hidden micro-state repair prevents it from qualifying as a macro-object under the declared constraints.

4.10 Reporting and Evidential Discipline

To keep conclusions comparable across cases, closure assessments should state five items explicitly.

1. Target partition and rationale.
2. Regime and horizon specification.
3. Admissible intervention class and justification.
4. Diagnostics used and their agreement or disagreement.
5. Final verdict category: robust, qualified, or indeterminate.

This disclosure structure matters because it prevents silent shifts in target, timescale, or admissibility from being mistaken for genuine ontological progress.

It also keeps the verdict vocabulary tied to ontology rather than to mere confidence. A robust verdict reports that a candidate occupies a stable place in the relevant closure structure under modest checks. A qualified verdict reports a fragile placement that can still support selective ontological commitment under explicit regime limitations. An indeterminate verdict reports that no stable placement has yet been secured. Rosas et al. provide the formal architecture for nested closure structure; this paper's contribution is to make that structure explicit in ontology reporting (Rosas et al. 2024).

Diagnostics do not replace criterion-level argument. They provide comparative evidence about whether a candidate satisfies the criterion under declared constraints. No single proxy is treated as infallible. Confidence rises when different diagnostics track the same rank-order among candidate partitions and when those rankings remain stable under modest perturbation tests.

Verdict categories are graded accordingly. Robust verdicts require convergence and stability. Qualified verdicts fit mixed but non-trivial evidence. Indeterminate verdicts fit persistent instability. The diagnostic set can vary by domain, but it should answer one fixed question: do within-class micro-differences still matter for the relevant macro-transition profile under the declared constraints? In practice, three checks are usually enough.

1. Leakiness proxy based on predictive gain from added within-class micro-features.
2. Within-class transition-profile divergence.
3. Intervention-response invariance under admissible perturbation.

Agreement across these checks increases warrant. Persistent disagreement is evidence for revision or downgrade, not for forced commitment. When interventional evidence is sparse or unavailable, the remaining diagnostics can still support qualified status, but robust objecthood requires that the interventional leg not be entirely missing.

Minimal triangulation recipe: require stability under modest horizon variation and under at least two defensible admissibility perturbations, while at least two diagnostics agree on candidate rank-order. If this condition fails, return qualified or indeterminate status rather than robust objecthood.

5 Objections and Replies

The following objections move from foundational worries to evidential-discipline worries, but the core answer remains stable: closure under fixed constraints supplies ontological discipline without demanding a single privileged descriptive level.

Three standing commitments apply throughout the replies below and will not be restated each time. First, the account does not deny microphysical completeness; it claims non-redundancy at the explanatory and interventional level for declared macro-targets. Second, all verdicts are indexed to the declared constraints, so objections that presuppose context-free ontological claims address a view the paper does not hold. Third, the strongest conclusion is conditional on structural realist and interventionist commitments; readers who withhold those commitments can still accept the narrower anti-gerrymandering result.

5.1 "Beyond Screening Off"

Objection: Franklin and Robertson already provide a principled admittance criterion for rainforest realism. If emergence requires screening off plus novelty, one might wonder what substantive work is left for closure to do (Franklin and Robertson 2024).

Reply: screening off and novelty are indispensable, but they are not sufficient. Franklin and Robertson's criterion asks whether the macro-variable makes micro-details conditionally irrelevant for the relevant explanandum while contributing a genuine macro-level dependency. Closure asks a further question: does the candidate macro-variable carry its own transition structure under admissible intervention, or was the screening off borrowed from elsewhere?

The common-cause case from §3.3 supplies the wedge: a proxy can inherit observationally genuine screening off from an upstream driver while failing to preserve its own transition structure under direct admissible perturbation. The disagreement is between dynamic observational screening off and transition stability under admissible perturbation.

The argument here relies on Franklin and Robertson's program. Their criterion removes obvious gerrymanders and clarifies the importance of law-like macrodependence. The claim is simply that one further filter is needed for robust admission to the rainforest, best understood as a completion of the admittance protocol introduced in §1.

5.2 "Instrumentalism and Admissibility"

Objection: target selection is interest-relative, so the view still collapses into usefulness.

Reply: target selection can be interest-shaped without making closure verdicts preference-shaped. Once regime, horizon, and admissibility are fixed, whether macro-transitions are autonomous is a system fact under explicit constraints.

The stronger version invokes Dennett's Martian (Dennett 1991, p. 42): if an ideal micro-predictor can forecast everything, macro realism looks optional. But this infers ontological redundancy from representational power. Even a micro-omniscient predictor coexists with objectively better macro-bottlenecks for target dynamics, and ignoring those bottlenecks increases representational burden without erasing macro transition structure.

Observer power changes convenience, not closure facts. Whether a partition is transition-autonomous under fixed constraints does not vary with who computes it.

Admissibility version of the same objection: admissibility criteria already encode what counts as the relevant level. The reply appeals to hierarchical order and disclosure rules. Admissibility is fixed by physically realizable, regime-preserving control channels before partition scoring. Post hoc admissibility revision triggers restart. This makes back-fitting explicit and sanctionable.

Residual disagreement can remain. The account does not promise universal convergence from one pass. It promises disciplined comparison and explicit downgrade when verdicts are unstable across nearby defensible admissibility specifications. This is a feature for skeptical readers: hard cases are flagged as unresolved rather than settled by stipulation.

Instrumentalist pressure hides in two forms: ontology should track predictive utility only, or ontology is unnecessary once predictive utility is available. Both are wrong. Utility without closure is too permissive, and closure without ontology understates what stable intervention-guiding structure provides. What closure adds is commitment discipline, not metaphysical extravagance. If a candidate repeatedly supports stable counterfactual control under fixed constraints, refusing any ontological standing starts to look like an empty verbal policy rather than a substantive alternative. Rather than forcing maximal realism, the paper forces explicit criteria for when anti-realism remains credible.

The strongest anti-instrumentalist point is structural rather than rhetorical. The objection that leakiness is "only" predictive error presupposes a gap between tracking causal structure and describing what is real. But under the paper's structural realist commitments, there is no further ontological layer for that gap to open onto. Stable counterfactual and predictive organization is not evidence for some deeper objecthood condition left unspecified elsewhere; it is what the relevant reality consists in at that grain. A high-leak partition is not one where we failed to see the real object hiding underneath. It is one where the candidate lacks autonomous transition structure. Leakiness therefore measures ontic deficiency in the candidate partition, not merely our ignorance. Once the test and the target are properly separated, the prediction miss is itself the structural fact.

The cheap-coding objection fails for the same reason. A recoding can improve local fit and still fail closure under admissible perturbation. The criterion is therefore representation-hostile in exactly the way permissiveness critiques demand.

5.3 "Causal Exclusion Still Defeats Macro Claims"

Objection: if microphysics is causally complete, macro-level causal claims are redundant (Kim 1998).

Reply: closure does not posit a second fundamental cause layer; it identifies when macro-description is sufficient for prediction and intervention at the relevant target level, leaving microphysical implementation untouched.

The standard reply to Kim's exclusion argument is that macro-causation is explanatorily indispensable. The closure criterion allows a stronger, structural reply: macro-states are non-redundant because they filter out causal degeneracy. Due to multiple realizability, myriad distinct micro-configurations map to identical macro-outcomes. Tracking every electron is not merely computationally expensive; it actively introduces non-difference-making noise into the causal model. When a system achieves closure, the macro-boundary strips out this micro-level degeneracy, effectively concentrating causal power. The macro-variable is therefore not a convenient shorthand competing with a fundamental micro-cause; it is a more informative causal parameter for the target outcome because it isolates the relevant difference-maker. Supervenience establishes a dependency relation, not an identity relation, and causal priority follows the level where conditional independence actually screens off this noise.

This reply also handles the fat-handedness worry that macro-interventions inevitably co-intervene on micro-states (Dewhurst 2021). Co-intervention on the micro-level is real, but closure ensures that the within-class micro-differences it produces do not change macro-transitions; the co-intervention is therefore non-difference-making for the macro target. Macro-level interventions can be successful and projectible across perturbations without denying microphysical realization (Woodward 2003; Pearl 2009; Hoel et al. 2013). Compatibility with microphysical completeness is a design feature, not a concession.

The overdetermination worry can be handled directly. The proposal does not claim two independent sufficient causes for one event. It claims one implemented process with multiple adequate descriptions at different explanatory and control targets. Exclusion pressure weakens once adequacy is indexed to target and intervention class rather than to a single privileged level.

Relative to Kim's strongest formulations, the key move is target-indexing rather than layer multiplication (Kim 1998, 2005). A closed macro-description can be non-redundant without competing for micro-level fundamentality, because non-redundancy is inferred when micro-detail ceases to add control-relevant information for the target outcome under fixed constraints. The claim concerns which counterfactual dependencies remain stable for a declared class of interventions. Compatibility with microphysical completeness by itself is too weak to do this work, because it leaves open whether macro-variables are dispensable shorthand. Closure narrows that space: when it holds, dispensability must be argued against a stated record of interventional and predictive sufficiency, not simply asserted.

5.4 "Autonomy, Pluralism, and Distinctiveness"

Objection: the view sounds like a formal restatement of familiar special-sciences autonomy claims, not a distinctive realism thesis.

Reply: the paper is continuous with autonomy insights but not equivalent to them. Generic autonomy claims often remain permissive about what qualifies as a level-worthy grouping; this view adds a discriminating closure condition with explicit exclusion and downgrade rules, and that addition changes verdict structure. The difference is both practical and philosophical: practical because the account gives a protocol for ruling out high-maintenance, transition-incoherent candidates; philosophical because it ties realism commitment to transition autonomy under admissible interventions rather than explanatory convenience alone. In this respect, causal-emergence style results are treated as evidence within a closure-governed approach, not as a replacement for the criterion itself (Hoel et al. 2013).

The same comparison can be made with the mechanisms literature. Mechanistic accounts are right to tie level talk to organized structure that supports stable intervention and explanation (Machamer et al. 2000; Craver 2007). Recent structuralist work on mechanisms makes the same pressure explicit: object-based metaphysics for mechanisms faces the problem of fitting special-science ontology to physics without collapsing it into mere convenience (Jiang 2024). But even that move does not by itself settle which variables or entities have been carved at the right grain for the target explanation. Closure addresses that earlier question by testing whether a candidate macro-variable screens off the micro-differences that would otherwise keep the mechanism explanatorily open.

The view should therefore be read as a constrained realism refinement of autonomy views, not as an unrelated alternative and not as a mere restatement. The conceptual difference can be stated directly. Generic autonomy claims assert that higher levels matter. Closure specifies *when* they matter, *how much* they matter (graded by leakiness), and *when the claim should be withdrawn* (explicit downgrade conditions). A realism criterion without withdrawal conditions is not a criterion at all. It is an assertion.

A more specific comparison is worth drawing. The classical arguments for the autonomy of the special sciences established that higher-level kinds can be explanatorily indispensable precisely because they abstract over heterogeneous micro-realizations (Putnam 1967; Fodor 1974). Multiple realizability is in fact a structural feature of any closed macro-kind: if a partition closes, different microstates within the same macroclass by construction produce the same macro-transitions, making the kind multiply realized in the relevant sense. Multiple realizability is therefore necessary for closure, but it is not sufficient: a purely extensional claim about kind membership says nothing about the dynamical coherence of the realizers. A disjunctive aggregate can have heterogeneous realizers while completely failing closure, because those realizers can have utterly divergent transition profiles. What closure adds is the dynamical requirement: the realizers must not only fall under the same label but must produce the same macro-transitions under the declared constraints. This gap, between what a kind groups together extensionally and how those instances behave dynamically, is precisely where gerrymandered kinds exploit multiple-realizability intuitions without earning macro-object status. The criterion formalizes the transition condition that the autonomy tradition always needed to separate genuine higher-level kinds from merely disjunctive aggregates (Shapiro 2000). The monetary illustration in §4.5 provides a concrete instance: coins, ledger entries, and digital balances are heterogeneous realizers, but what earns money macro-object status is not the heterogeneity of realizers but the transition-equivalence of those realizers under admissible financial interventions.

Pluralism version of the objection: if multiple grains can satisfy closure, ontology becomes permissive again. The reply is that plurality of closed grains does not imply arbitrariness. Once regime, horizon, and admissibility are fixed, which partitions close is determined by transition structure, not by analyst choice. When multiple candidates remain viable, robustness and minimality rank them as an objective relation among candidates under fixed constraints. In many domains these relations are further structured by coarse-graining order: when one closed partition is a refinement of another, their relation is nesting, not rivalry, which disciplines pluralism rather than dissolving it.

Andersen's density-of-structure point helps explain why this is not an embarrassment (Andersen 2025). If, as he puts it, "the world may be dense with structure," then one should expect more than one admissible closed grain in some systems (Andersen 2025, p. 1). The philosophical task is not to force uniqueness where the dynamics do not supply it. It is to report where in the closure space each candidate stands, how stable that standing is, and which grains are more robustly supported under the declared constraints.

This is a virtue for complex systems, not an embarrassment. Forcing a single grain in every context would be a stronger and less defensible claim than the paper needs.

5.5 "The Markov Template is Too Restrictive"

Objection: strong lumpability presupposes first-order Markov microdynamics and excludes memory-dependent systems.

Reply: the criterion is transition sufficiency, not first-order Markovity. Strong lumpability is an exact benchmark. In memory-bearing systems, state can be enriched with relevant history and the same closure question can be asked over that enriched state. This changes state representation, not criterion content, and the basic point has already been built into the exact benchmark discussion in Section 3.

Concrete cases make this less abstract. Immune response modeling often depends on historical exposure profiles, and institutional dynamics often depend on path-dependent enforcement and trust trajectories. In both cases, a memoryless state is too thin, but a minimally history-enriched state can still support closure testing at the target grain.

Minimality still matters. Enrichment should be the least extension needed to preserve closure performance under fixed constraints. The account also accepts an important edge case: if the minimally sufficient enriched state approaches micro-level complexity, then closure may be recovered without meaningful compression, which signals that no informative macro-partition has been found at the target grain rather than that robust macro-objecthood has been achieved.

5.6 "Formal Closure Collapses the View into Formalism"

Objection: formal systems are closed by stipulation, so closure cannot ground empirical objecthood.

Reply: stipulated and induced closure must be separated. Formal systems can have objective internal closure under constitutive rules. The present criterion concerns induced closure in implemented spatiotemporal dynamics. Internal formal coherence does not by itself settle empirical macro-objecthood claims.

Formal systems are invented, but not freely. What counts as intelligible formal practice is constrained by stable identity conditions, compositional inference, and coherent consequence. Those constraints explain why formal work can guide empirical reasoning without itself deciding empirical objecthood.

No collapse follows. The paper's argument is narrower and clearer: induced closure is the objecthood criterion for implemented macro-patterns in regime.

5.7 "Closure Is Too Strong and Eliminates Most Special Sciences"

Objection: if closure requires that within-class micro-differences not matter for macro-transitions, then most special-science kinds will fail. Biological species, psychological states, and economic categories are notoriously leaky (Boyd 1991). The criterion eliminates the ontology it was supposed to discipline.

Reply: the objection conflates robust objecthood with any objecthood verdict at all. Three verdict categories are available, not one. Robust objecthood requires stable closure under fixed constraints. Qualified objecthood fits cases where closure is partial, regime-limited, or supported by convergent but incomplete evidence. Indeterminate status fits cases where evidence is too unstable to warrant commitment.

Most special-science kinds fall into the qualified range, and that is the correct result. Biological species exhibit substantial transition autonomy within ecological and evolutionary regimes while leaking at boundary cases (hybrid zones, ring species, horizontal gene transfer). Psychological kinds often support predictive and interventional regularity within clinical or experimental regimes while failing under broader perturbation. The criterion does not eliminate these kinds. It gives them the status their closure evidence supports: qualified objecthood with explicit regime limitations, rather than either full robust status or wholesale elimination.

Nearby real-pattern work in biology provides a useful foil. One can characterize biological objects by their explanatory power and pragmatic utility, and by the availability of multiple valid carvings, without yet having shown that lower-level variation is transition-irrelevant in the stronger sense used here (Jiang 2025). Such utility matters, but it does not by itself settle objecthood.

The objection therefore proves too much. A criterion that granted robust objecthood to every special-science kind regardless of leakiness would be the permissive view this paper is designed to replace. The gain from graded verdicts is that they match evidential reality rather than forcing a binary choice between full realism and eliminativism.

5.8 "Predictive Success and Failure Conditions"

Objection: a model can predict well without warranting ontological commitment.

Reply: that is correct, and the account does not deny it. Predictive success functions as evidence only when it aligns with closure diagnostics and interventional stability.

This is also the natural place to distinguish the view from the strongest nearby tightening of real-pattern realism. Millhouse shows why compression, similarity, and counterfactual support already improve on a purely permissive reading of Dennett (Millhouse 2022). But even those stronger constraints leave open whether the candidate partition itself carries autonomous transition structure under admissible perturbation.

Two consequences follow. Diagnostics operationalize what the argument says should matter, but they do not replace criterion-level reasoning. And the criterion is falsifiable in its own terms: if candidate partitions repeatedly fail closure diagnostics across defensible regimes and admissibility classes, commitment should be withdrawn or downgraded. A proposal that cannot risk downgrade is not a serious realism criterion.

5.9 "Horizon-Relativity Makes the Criterion Too Weak"

Objection: if closure is indexed to horizon, any candidate can be made to pass at a short enough horizon.

Reply: horizon indexing is a constraint, not an escape clause. Two requirements prevent it from becoming one.

First, horizons must be anchored in independently identified timescale structure in the target domain: known relaxation windows, intervention latency, control-response periods, or similar physically motivated temporal features. A horizon chosen only because it makes a favored partition look closed has no independent standing.

Second, verdicts must survive a modest-extension stability check. A candidate that passes at horizon L but fails immediately at $L + \Delta$ for small, defensible Δ receives, at best, qualified status. Robust objecthood requires that closure not be razor-thin in temporal scope.

Together, these requirements force explicit timescale discipline and prevent silent switching between incompatible temporal targets. If robustness fails, the verdict downgrades rather than the criterion breaking: the claim was too strong for the evidence.

5.10 "The Criterion Is Too Conservative"

Objection: by demanding closure and downgrade discipline, the criterion may be too conservative and may miss emerging macro-objects.

Reply: conservatism is partly intentional. The paper aims to avoid premature ontological inflation. Still, the criterion is not rigid. It allows qualified verdicts for emerging patterns when closure evidence is partial but improving, and it allows verdict revision as regimes stabilize and intervention evidence accumulates.

The strongest version of the objection should also be granted. Some good macro-kinds enter science before robust intervention evidence is in hand. Early thermodynamic kinds are the obvious sort of case. The present proposal should not be read as saying that such kinds were therefore unreal until a later evidential stage. The better verdict is often qualified: strong enough for selective commitment because the pattern is doing non-trivial explanatory work, but not yet strong enough for unqualified closure-based admission.

So the criterion does not require all-or-nothing maturity before any commitment. It requires that commitment strength track available closure evidence. Woodward (2021) similarly argues that while minimal interventionist criteria establish causality, invariance is a graded virtue. Our criterion of closure can be seen as the dynamical realization of this: a macro-object is "proportional" precisely when it minimizes the omission of relevant micro-detail. This is a virtue for dynamic systems where objecthood can be developmental rather than instantaneous.

Compression gains may therefore appear before robust closure is demonstrated; the account treats this as evidential lead, not ontological conclusion. Commitment can strengthen as closure evidence accumulates, and should remain qualified or be withdrawn when it does not.

6 Conclusion: A Disciplined Realism Claim

Rainforest realism is compelling, but its admittance criteria remain incomplete. The view defended here is cumulative: compression identifies candidate patterns; screening off plus novelty, in Franklin and Robertson's sense, removes obvious gerrymanders and secures serious macrodependence (Franklin and Robertson 2024); and closure then determines which survivors carry autonomous transition structure under fixed regime, horizon, and admissible intervention class. Strong lumpability provides the exact benchmark. Leakiness and convergent diagnostics extend the same criterion to non-ideal settings without changing its content.

This further step sharpens the non-redundancy demand already implicit in the Primacy of Physics Constraint (§2.4): a candidate is non-redundant when lower-level variation within its macroclasses stops mattering to the relevant macro-transition profile under declared constraints. In the recent literature, the criterion supplies what Franklin and Robertson's screening-off test, Rosas et al.'s formal diagnostics, Ladyman and Lorenzetti's effective realism, and recent dynamic formulations of pattern realism each leave open: an explicit admittance and downgrade discipline for macro-objecthood (Franklin and Robertson 2024; Rosas et al. 2024; Ladyman and Lorenzetti 2024; Wallace 2024; Ladyman 2026).

The proposal supports selective commitment: robust where closure is stable, qualified in borderline cases, explicitly downgraded where verdicts depend on fragile admissibility or horizon choices. It is intentionally middle-range, stronger than compression-only pattern realism because it adds exclusion and downgrade structure, weaker than a total-level metaphysics because it is restricted to induced closure in spatiotemporal systems under declared constraints.

The positive metaphysical payoff is correspondingly more explicit than a generic many-level realism. Once closure governs admission, the rainforest is best understood as the structured space of computationally closed coarse-grainings, and the verdict categories make that structure reportable: robust where placement is stable, qualified where ontological support is real but fragile, and indeterminate where no stable placement has yet been secured.

The view separates itself from eliminativist pressure by neither treating higher-level descriptions as fictions by default nor granting them standing by convenience alone. The test is whether they track stable transition structure under admissible intervention. If they do, realist commitment follows on the paper's stated terms. If they do not, realism should be downgraded rather than protected by rhetoric.

The proposal is conditional on the structural realist and interventionist commitments stated in §1; readers who withhold those commitments can still accept the narrower anti-gerrymandering result. It is not a complete metaphysics of levels and not a universal methods manual. Its contribution is a philosophical criterion with application discipline. Methodological implementation details remain downstream of this argument, and refining them does not alter the criterion's philosophical content. Future work should test the staged admittance protocol across domains by comparing candidate partitions under fixed constraints, rather than by multiplying new concepts. The paper has tried to make its commitments explicit and to address worries about circularity, instrumentalist drift, and scope ambiguity directly. If those replies fail, they fail on the merits of the stated argument rather than because those issues were left silent. One further question, left for future work, is whether this closure-based account of realist commitment also has implications for substrate-independence or more abstract forms of structural objecthood.

References

Ainsworth PM (2009) Newman's objection. The British Journal for the Philosophy of Science 60(1):135–171. https://doi.org/10.1093/bjps/axn051

Andersen HK (2025) The density of structure. Synthese 206(5):237. https://doi.org/10.1007/s11229-025-05305-y

Ando A, Simon HA (1961) Aggregation of variables in dynamic systems. Econometrica 29(2):111–138. https://doi.org/10.2307/1909285

Batterman RW (2002) The Devil in the Details: Asymptotic Reasoning in Explanation, Reduction, and Emergence. Oxford University Press, New York. https://doi.org/10.1093/0195146476.001.0001

Boyd R (1991) Realism, anti-foundationalism and the enthusiasm for natural kinds. In: Philosophical Studies, vol 61. p 127–148. https://doi.org/10.1007/BF00385837

Cover TM, Thomas JA (2006) Elements of Information Theory, 2nd edn. Wiley-Interscience, Hoboken, NJ. https://doi.org/10.1002/047174882X

Craver CF (2007) Explaining the Brain: Mechanisms and the Mosaic Unity of Neuroscience. Oxford University Press, Oxford. https://doi.org/10.1093/acprof:oso/9780199299317.001.0001

Dennett DC (1991) Real patterns. The Journal of Philosophy 88(1):27–51. https://doi.org/10.2307/2027085

Dewhurst J (2021) Causal emergence from effective information: Neither causal nor emergent? Thought: A Journal of Philosophy 10(3):158–168. https://doi.org/10.1002/tht3.489

Fodor JA (1974) Special sciences (or: The disunity of science as a working hypothesis). Synthese 28(2):97–115. https://doi.org/10.1007/BF00485230

Franklin A, Robertson K (2024) Emerging into the rainforest: Emergence and special science ontology. European Journal for Philosophy of Science 14(4):90. https://doi.org/10.1007/s13194-024-00622-4

Goodman N (1983) Fact, Fiction, and Forecast, 4th edn. Harvard University Press, Cambridge, MA. Originally published 1955

Haugeland J (1998) Having Thought: Essays in the Metaphysics of Mind. Harvard University Press, Cambridge, MA

Hoel EP, Albantakis L, Tononi G (2013) Quantifying causal emergence shows that macro can beat micro. Proceedings of the National Academy of Sciences 110(49):19790–19795. https://doi.org/10.1073/pnas.1314922110

van Inwagen P (1990) Material Beings. Cornell University Press, Ithaca, NY

Jiang Y (2024) The metaphysics of mechanisms: An ontic structural realist perspective. Synthese 204:32. https://doi.org/10.1007/s11229-024-04684-y

Jiang Y (2025) Biological object as real patterns: Reconciling processualism and scientific realism. Synthese 206(4):185. https://doi.org/10.1007/s11229-025-05285-z

Kemeny JG, Snell JL (1960) Finite Markov Chains. Van Nostrand, Princeton, NJ. Reprinted 1976, Springer

Kim J (1998) Mind in a Physical World. MIT Press, Cambridge, MA

Kim J (2005) Physicalism, or Something Near Enough. Princeton University Press, Princeton, NJ

Ladyman J (2026) Patterns all the way up: Prolegomena to a future naturalised metaphysics. In: Millhouse T, Petersen S, Ross D (eds) Dennett's Real Patterns in Science and Nature. MIT Press, Cambridge, MA. Preprint circulated in 2024; final version in edited volume

Ladyman J, Lorenzetti L (2024) Effective ontic structural realism. The British Journal for the Philosophy of Science https://doi.org/10.1086/729061

Ladyman J, Ross D (2007) Every Thing Must Go: Metaphysics Naturalized. Oxford University Press, Oxford

Landauer R (1961) Irreversibility and heat generation in the computing process. IBM Journal of Research and Development 5(3):183–191. https://doi.org/10.1147/rd.53.0183

Lewis D (1986) On the Plurality of Worlds. Blackwell, Oxford

Machamer P, Darden L, Craver CF (2000) Thinking about mechanisms. Philosophy of Science 67(1):1–25. https://doi.org/10.1086/392759

Millhouse T (2022) Really real patterns. Australasian Journal of Philosophy 100(4):664–678. https://doi.org/10.1080/00048402.2021.1941153

Millhouse T, Petersen S, Ross D (eds) (2026) Dennett's Real Patterns in Science and Nature. MIT Press, Cambridge, MA. https://doi.org/10.7551/mitpress/15550.001.0001

Pearl J (2009) Causality: Models, Reasoning, and Inference, 2nd edn. Cambridge University Press, Cambridge

Putnam H (1967) Psychological predicates. In: Capitan WH, Merrill DD (eds) Art, Mind, and Religion. University of Pittsburgh Press, Pittsburgh, p 37–48

Rosas FE, Geiger BC, Luppi AI, Seth AK, Polani D, Gastpar M, Mediano PAM (2024) Software in the natural world: A computational approach to hierarchical emergence. arXiv preprint arXiv:2402.09090. https://doi.org/10.48550/arXiv.2402.09090, URL https://arxiv.org/abs/2402.09090

Ross D (1995) Real patterns and the ontological foundations of microeconomics. Economics and Philosophy 11(1):113–136. https://doi.org/10.1017/S0266267100003242

Shalizi CR, Crutchfield JP (2001) Computational mechanics: Pattern and prediction, structure and simplicity. Journal of Statistical Physics 104(3-4):817–879. https://doi.org/10.1023/A:1010388907793

Shapiro LA (2000) Multiple realizations. The Journal of Philosophy 97(12):635–654. https://doi.org/10.2307/2678460

Simon HA (1962) The architecture of complexity. Proceedings of the American Philosophical Society 106(6):467–482. https://doi.org/10.2307/985254

Wallace D (2024) Real patterns in physics and beyond. URL https://philsci-archive.pitt.edu/id/eprint/23888/, preprint

Wimsatt WC (1981) Robustness, reliability, and overdetermination. In: Brewer MB, Collins BE (eds) Scientific Inquiry and the Social Sciences. Jossey-Bass, San Francisco, p 124–163

Wimsatt WC (2000) Emergence as non-aggregativity and the biases of reductionisms. Foundations of Science 5(3):269–297. https://doi.org/10.1023/A:1011342202830

Woodward J (2003) Making Things Happen: A Theory of Causal Explanation. Oxford University Press, Oxford

Woodward J (2015) Methodology, ontology, and interventionism. Synthese 192(11):3577–3599. https://doi.org/10.1007/s11229-014-0479-1

Woodward J (2021) Causation with a Human Face: Normative Theory and Descriptive Psychology. Oxford University Press, Oxford. https://doi.org/10.1093/oso/9780197585412.001.0001